

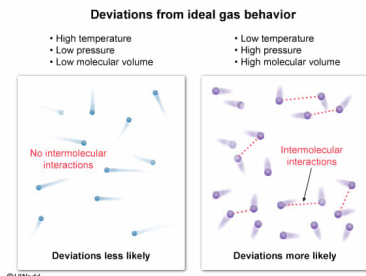
A scientist uses extreme temperature environments to study the non-ideal behavior of gas molecules. One way to increase the non-ideal behavior of gas molecules under investigation would be to:

- ☐ A. use gas molecules with a smaller molecular volume. [4%]
- ☐ B. decrease the pressure within the gas molecule container. [12%]
- ☒ C. use gas molecules with more significant intermolecular interactions. [73%]
- ☐ D. transfer large quantities of heat to the gas molecule container. [10%]

Correct

72% answered correctly

Explanation:



The first assumption made when modeling gas systems is that the gas molecules are **ideal**, with negligible molecular volume and no intermolecular forces. However, many real gas molecules do exert intermolecular forces on one another, and although molecular volume is very small, it is not zero. As a result, real gases exhibit **deviations from ideal behavior**.

Attractive **intermolecular interactions** draw gas molecules toward the center of the container, decreasing the frequency of collisions with the container wall. Consequently, the ideal gas pressure (P_i) is always greater than the real pressure (P_r). This discrepancy is corrected by a term relating an attraction constant (a) to the square of the gas molecule density (n/V), which is the moles of gas (n) per unit of volume (V):

$$P_i = P_r + a \frac{n^2}{V^2}$$

Because every gas molecule occupies a **non-negligible volume** that diminishes the space through which the gas can move, ideal gas volume (V_i) is always less than real gas volume (V_r). This discrepancy is corrected by a term relating moles of gas (n) to a volume constant (b):

$$V_i = V_r - nb$$

The van der Waals equation is a restatement of the **ideal gas equation** with the correction terms added:

$$\left(P_r + a \frac{n^2}{V^2} \right) (V_r - nb)$$

The second assumption made when modeling gas systems is that gas molecules move freely throughout the system. However, **high-pressure** systems force gas molecules into close proximity, increasing the frequency of intermolecular interactions. Furthermore, **low-temperature** systems reduce the **kinetic energy** of gas molecules, decreasing the probability of molecules "escaping" intermolecular interactions.

Therefore, deviations from ideal gas behavior are most often observed when gas molecules with non-negligible volume and significant intermolecular interactions are placed under conditions of high pressure and low temperature (**Choice C**).

Conversely, non-ideal behavior of a real gas *decreases* when:

- Molecular volume is small, approximating the ideal model (**Choice A**).
- Decreases in pressure diminish the frequency of intermolecular interactions (**Choice B**).
- Increases in temperature diminish the longevity of intermolecular interactions (**Choice D**).

Educational objective:

Deviations from ideal gas behavior are observed when gas molecules with non-negligible volume and significant intermolecular interactions occupy a gas system at high pressure and low temperature.

Time spent: 0 sec
Subject:
General Chemistry

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System:
Thermodynamics, Kinetics & Gas Laws